AgriVision Research Paper: Team Context

Everything decided so far, all deadlines, what must be done and how. Written 4 Nov 2026 for Navyashree B H and the team.

RULE: This document is INTERNAL. It names people and the college. The paper itself must be anonymous (double-blind). Never copy names, college, guide, logos or GitHub links into the paper or its figures.

1. The goal in one paragraph

We are writing a research paper on our IoT agricultural rover (AgriVision) for **CompSIF-2027** (3rd International Conference on Computing for Sustainability and Intelligent Future, BMS Institute of Technology and Management, Bengaluru, 12-13 March 2027). Accepted papers are submitted for possible inclusion in IEEE Xplore. Best-fit track: **Track 2, AI for a Sustainable Future** (backup: Track 5, Embedded Systems). The paper must show a working rover plus honest, measured evidence for the computer-vision part.

2. Deadlines (all times AoE, Anywhere on Earth)

Date	What	Notes
14 Nov 2026, 23:59	**Paper submission (firm)**	Microsoft CMT: cmt3.research.microsoft.com/CompSIF2027. We plan to submit on the evening of 13 Nov
10 Nov 2026 (internal)	Evidence handover: all CV results, figures, CSVs to Shreyas	Gives 4 days to write and polish
1 Jan 2027	Acceptance notification	
15 Jan 2027	Camera-ready paper + signed IEEE copyright form	We may add new results here (e.g. 4G module, more field data)
25 Jan 2027	Early-bird registration closes	Join IEEE (optional) before this
20 Feb 2027	Final registration	
12-13 Mar 2027	Conference, in person, Bengaluru	At least one author must register and present; live rover demo planned

As of 4 Nov there are 10 days left to submit.

3. Submission rules we must obey

- **Format:** official IEEE conference template, double column, PDF only.
- **Length:** research paper max **6 pages including figures, tables and references**. Alternative: 4-page work-in-progress / poster category. Papers lacking depth may be moved there. (Our 6-vs-4 decision is still open; the aim is 6 if the evidence is ready.)
- **Double-blind:** no author names, affiliations, acknowledgements or identifying details. Cite our own earlier work in the third person.
- Format, length or anonymity violations lead to rejection without review.
- No posting the paper on arXiv or similar sites (desk rejection).
- Author list is final at submission: no additions, removals or reordering afterwards.
- Original, unpublished work only. IEEE plagiarism screening applies.
- One author must register and present in person; each accepted paper needs its own registration and a signed IEEE copyright form.

4. Team and roles

Name	Role (proposed)
Navyashree B H	Computer vision lead: ESP32-CAM + pan-tilt circuit, datasets, model training, results, CV literature (see Computer Vision folder)

Name	Role (proposed)
Shreyas R S	Rover/IoT software and firmware (E:\myrover), paper assembly and writing, submission
Sumaiya Firdoes	To be assigned (suggestions: field data collection, sensor measurements, figures)
Tejaswini D	To be assigned (suggestions: literature table, link-latency and 4G tests, proofreading)
Guide: Ms. Sathyabhama R	Approval before submission; her place on the author list is still to be decided

Author order (decided): alphabetical by first name: 1 Navyashree B H, 2 Shreyas R S, 3 Sumaiya Firdoes, 4 Tejaswini D. Still open: whether the guide is listed as an author, and who the **presenting author** is (the person who registers and travels to Bengaluru). Names go only into the CMT submission form. Registration fees (from the site): student non-IEEE Rs 5,500 early bird (Rs 6,500 standard); student IEEE member Rs 4,500 / 5,500; attendee or accompanying person Rs 1,000 / 1,500. Whether all four can attend and present is not stated; email compsif@bmsit.in (phone +91-8026146823) after acceptance.

5. The project as we will describe it

Motto: farmers cannot continuously monitor their land (especially smallholders in India). A rover monitors the field and the farmer watches from a nearby place such as a farm house.

Base codebase: `E:\myrover` (the newest, leanest version; everything in it is working). Other folders (agrirover/new-design, cv plants, crop-recommender) are older versions: we reuse ideas such as the plant detector, but we do not describe features that are not in the final build. The crop-recommender is dropped (weak data, only a training accuracy).

Part	Final decision
Motor driver	L298N (single H-bridge board)
Controllers	Driver ESP32, separate fire-sensor ESP32, ESP32-CAM on pan-tilt servos
Confirmed sensors	Soil moisture, rain, light sensor (automatic front and rear lights), 5-channel flame sensor
Temperature/humidity	DHT-series sensor kept for the prototype (cost). Weatherproof BME280/SHT31 and a waterproof soil-temperature probe are mentioned only as future work for a field-deployable version
Obstacle sensing	4 ultrasonic sensors, front one on a servo (assumed front, left, right, rear). Goal: automatic obstacle avoidance. We claim reactive avoidance and row-centering only; full path planning is future work
Camera	ESP32-CAM on a 2-servo pan-tilt (pan GPIO 13, tilt GPIO 14), own 5 V servo supply, common ground
Software	Node.js server with WebSocket hub, React dashboard, auth, MongoDB run history; link-quality probes (RTT, loss, WiFi RSSI)
Long range	Goal up to 2 km from farm house to field by March via a 4G modem and a cloud server. Module NOT yet owned (see section 8)

Computer-vision pipeline (the paper's main contribution)

1. Stage 1, plant detector (YOLOv8n, one class): is there a plant, and where?
2. The rover approaches that plant and the pan-tilt camera takes close-ups (approach behaviour is planned)
3. Stage 2, leaf-health classifier (MobileNetV3-Small, tomato first): healthy or diseased, and which disease
4. If diseased: an alert on the dashboard with time, class, confidence, location and a damage estimate (location and damage % are planned: they need GPS or dead reckoning and a severity method)
5. Optional later: treatment (spraying). We treat this as future work

> **TIP:** We claim 'multi-crop capable, tomato-focused', never 'universal for all plants'. Lab-trained plant-disease models usually lose accuracy on real field photos, so we deliberately measure that gap and report it. That honesty is a strength.

6. Honesty rules (the most important section)

1. **Never invent or round up a number, result or reference.** Every result comes from a log, CSV or output file.
2. Use status words exactly: **Implemented** (works today, in E:\myrover), **In progress**, **Planned**, **Future work**. Planned work is allowed in the paper only when labelled; never write it in the present tense as if it works.

3. **~70% of the paper is built and measured, ~20% in-progress vision, ~10% roadmap.** Reviewers score technical depth; too much 'we will' turns it into a proposal.
4. Every reference must be opened and verified by a person (publisher page or DOI). AI tools can invent papers.
5. Split data **by plant, never by image**, or accuracy will look falsely perfect.
6. Keep raw outputs, logs and versions so any number can be reproduced.
7. Anonymity: no names, college, logos, farm names or faces in the paper or its figures.

7. What is already done

- Paper first draft: `Paper Versions/firstdraft.pdf` (3 pages, IEEE-style, anonymised). Unmeasured items are shown in red [brackets] and a red banner that must be removed before submission.
- `Computer Vision/` folder for Navyashree: guides 00-08 (start, concepts, hardware, dataset, training, results, checklist, literature, AI prompts), Starter Code (capture, split, train, evaluate, latency) and CSV Templates.
- Everything is committed and pushed to a GitHub repo (check that it is set to private). **Do not share the repo link with the conference.**
- Eight classic references are in the draft but still need verification: Mohanty 2016, Hughes and Salathe 2015, PlantDoc 2020, Ferentinos 2018, Kamilaris 2018, YOLO 2016, Ultralytics YOLOv8, MobileNetV3 2019.
- E:\myrover already provides measurable system facts: 100 ms drive-command cadence, 500 ms failsafe, 2 s round-trip probes, flame-sensor debounce 150 ms / 600 ms, 1.5 s light-sensor hold.

8. What still has to be done (with owners and dates)

The Computer Vision guide was dated from 1 Nov. Because today is 4 Nov, anything scheduled for 1-3 Nov is now due immediately. The 10 Nov handover date does not move.

#	Task	Owner	Due
1	Read Computer Vision guides 00-08; flash stock CameraWebServer; screenshot stream	Navyashree	5 Nov
2	Build pan-tilt circuit; run camera and pan-tilt tests (CSV logs, photos)	Navyashree	6 Nov
3	Download PlantVillage tomato + PlantDoc; create class-name mapping; DATA_SOURCES.txt	Navyashree	5 Nov
4	Collect own tomato images with plant IDs (field if access, else potted plants); expert label check	Navyashree + team	5-8 Nov
5	Train classifier (3 seeds) and detector; evaluate in-domain, cross-domain, ESP32-CAM set	Navyashree	6-9 Nov
6	Latency tests (camera snapshot, inference); figures; evidence folder	Navyashree	9-10 Nov
7	Rover measurements: WiFi RTT/loss/RSSI vs distance, camera frame time, battery runtime, ultrasonic avoidance trial, fire/soil/rain sanity tests	Shreyas + team	5-10 Nov
8	4G test: order SIM7600-class module (see below) OR run the rover through a phone 4G hotspot to a cloud server, and label it 'hotspot' honestly	Shreyas	order now
9	Literature table: 15-20 verified references (IoT, agri robots, disease DL, lightweight models, links)	Tejaswini + Navyashree	9 Nov
10	Fill paper placeholders, write results/discussion, check 6-page limit and anonymity, proofread	Shreyas + team	10-13 Nov
11	Guide approval of final PDF; decide guide author placement and presenting author	Team + guide	12 Nov
12	Submit on CMT with all four authors in the final order; save the confirmation e-mail	Shreyas	13 Nov evening

4G module shopping notes

- Buy a **4G LTE** modem breakout with UART at 3.3 V: SIMCom **SIM7600I** (India version), **SIM7600E-H** or **SIM7600G-H**; cheaper alternative **A7670C**. Needs Indian bands (B3, B5, B40; B41/B1/B8 a bonus).

- Avoid 2G-only modules (SIM800L, SIM900) and NB-IoT-only modules.
- Also needed: LTE antenna, an active Jio or Airtel data SIM tested in a phone first, a separate 5 V 2-3 A supply (buck converter) and a 1000 uF capacitor: the modem draws up to about 2 A in bursts and our rover already suffers brownouts.
- Check prices, band lists and stock yourself on the seller page: these were recommendations, not verified live listings.

Mandatory evidence if we want a 6-page paper

[ ] Detector precision, recall, mAP50, mAP50-95 and inference time
[ ] Classifier accuracy and macro-F1 on: in-domain, cross-domain (PlantDoc), own ESP32-CAM images; 3 seeds
[ ] Link measurements: RTT, loss, snapshot delivery (WiFi, and 4G or hotspot)
[ ] Real photos of the mounted camera and circuit; example detections including failures
[ ] At least one ablation (public data only vs public + our own images)

WATCH OUT: If evidence is not ready by 10 Nov: state the part as 'in progress' in the paper, use the 4-page work-in-progress category if needed, and add the missing results at camera-ready (15 Jan). Never fill a gap with a guessed number.

9. How the paper is organised (what goes where)

Section	Content	Evidence source
I Introduction	Problem (farm monitoring), gap, 4 contributions	Motto, related-work gap
II Related work	Plant-disease DL, lightweight models, IoT and agri robots	Literature table
III System architecture	Nodes, sensors, software, link, Table I status	E:\myrover, hardware photos
IV Vision pipeline	Camera node, detector, classifier, alert record	Navyashree's training logs
V Evaluation protocol	Datasets, plant-level split, metrics, three test sets	dataset_log, split counts
VI Results	Tables II (detector), III (classifier), IV (links)	CSVs from the evidence folder
VII Limitations and future work	Ultrasonic limits, sensor grade, domain gap, no spraying	Honest notes
VIII Conclusion	Two sentences of measured findings, then remaining work	Final results

10. Open questions (to settle this week)

[ ] Who is the presenting author (one person who registers and travels)?
[ ] Is the guide listed as an author or only approving? (Order cannot change after submission.)
[ ] Do we have access to a tomato field this week? If not, use potted plants and public data, and say so.
[ ] 6-page paper or 4-page work-in-progress? (Decide by 8 Nov based on evidence.)
[ ] Which 4G route: buy a modem now, or use the phone-hotspot path for the 14 Nov paper and upgrade in January?
[ ] Confirm the 4 ultrasonic layout: front on servo, left, right, rear.
[ ] Does the DHT sensor appear in the final wiring, and on which ESP32 (pins are tight)?
[ ] IEEE student membership: optional, saves about Rs 1,000 on the author fee only if the membership costs less. Not needed to submit.

11. Where things are

What	Where
First paper draft	Paper Versions/firstdraft.pdf
Computer-vision guide, code, templates	Computer Vision/ (start with 00_Start_Here_Navyashree.pdf)
Rover code (canonical)	E:\myrover

What	Where
Older versions and models	C:\Users\HP\Desktop\agrirover\new-design, cv plants, crop-recommender
Source documents, synopsis	docs/ (newsyn.pdf, synopsis.pdf)
Conference page	bmsit.ac.in/compsif (author guidelines, fees); previous proceedings on IEEE Xplore
Contact for questions	compsif@bmsit.in, +91-8026146823

One last rule: when in doubt, ask. A question costs two minutes; a wrong claim in a submitted paper cannot be fixed after 14 November.